

PG-DESD - C Programming Twisters Activity

1. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    char ch = 65;
    int x = 9;
    printf("%d, %d,", sizeof(ch), sizeof('A'));
    printf("%d, %d, ", sizeof(65), sizeof(++x));
    printf("%d", x);
    return 0;
}
```

2. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    int a = 3, b = 6;
    printf("%d, ", a = b);
    printf("%d, ", a == b);
    printf("%d, ", a != b);
    printf("%d, ", a=!b);
    return 0;
}
```

3. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    int a = -10, b = 3, c = 0, d;
    d = a++ || ++b && c++;
    printf("%d, %d, %d, %d, ", a, b, c, d);
    a = -10, b = 3, c = 0;
    d = c++ && ++b || a++;
    printf("%d, %d, %d, %d", a, b, c, d);
    return 0;
}
```

PG-DESD - C Programming Twisters Activity

4. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    int a = (1, 2, 3);
    int b = (++a, ++a, ++a);
    int c = (b++, b++, b++);
    printf("\n%d, %d, %d", a, b, c);
    return 0;
}
```

5. What will be the output of following program?

```
#include <stdio.h>
int main()
{
    printf("%d", ++4);
    return 0;
}
```

6. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    int a = 3, b = 6;
    if(a = b)
        printf("%d, %d", a, b);
    else
        printf("%d != %d", a, b);
    return 0;
}
```

7. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    int a = 3, b = 6;
    printf("%d ", a != b ? 0 : b : a : b);
    return 0;
}
```

PG-DESD - C Programming Twisters Activity

1. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    int k=1;
    switch(k++)
    {
        default : printf("0");
        case 1: case 2: printf("2");
        case 3*0: printf("1");
    }
    return 0;
}
```

2. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    float i=1.5;
    switch((int)i)
    {
        case 1: printf("1"); case 2: printf("2");
        default : printf("0");
    }
    return 0;
}
```

3. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    int a =3;
    while(a--)
    {
        printf("%d, ", a);
    }
    return 0;
}
```

4. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    int a = 1000000L;
    for(a=1; a<10; ++a);
    {
        printf("%d, ", a);
    }
    return 0;
}
```

5. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    char i = 0;
    do
    {
        printf("%d, ", i);
    }while(i--);
    return 0;
}
```

6. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    int i, j;
    for(i = 1; i <= 3; i+=2)
        for(j = 1; j <= 3; j+=2)
            printf(" %d, ", i+j);
    return 0;
}
```

PG-DESD - C Programming Twisters Activity

1. What will be the output of following program?

```
#include <stdio.h>
int x = 3;
int main(void)
{
    int x = 4, i;
    for(i = 0; i < x; ++i)
    {
        int x = 0; printf("%d ", x++);
    }
    return 0;
}
```

2. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    int main = 3;
    printf("%o", main);
    return 0;
}
```

3. What will be the output of following program?

```
#include <stdio.h>
int main = 3;
int main(void)
{
    printf("%o", main);
    return 0;
}
```

4. What will be the output of following program?

```
#include <stdio.h>

int num;
int function(int n)
{
    int num = 10;
    return num;
}
```

PG-DESD - C Programming Twisters Activity

```
int main(void)
{
    printf("%d %d\n", num, function(20));
    return 0;
}
```

5. What will be the output of following program?

```
#include <stdio.h>
```

```
int no1 = 71, no2 = 17;
void swapping(int no1, int no2)
{
    int temp = no2;
    no2 = no1;
    no1 = temp;
}
int main(void)
{
    int no1 = 17, no2 = 71;
    printf("%d %d ", no1 , no2);
    swapping(no1, no2);
    printf("%d %d\n", no1, no2);
    return 0;
}
```

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1. What will be the output of following program?

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    int a = 1;
```

```
    int *p = &a;
```

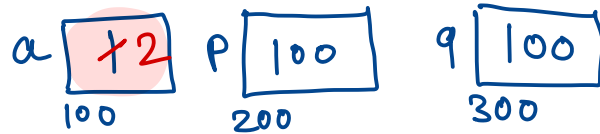
```
    int *q = p;
```

```
    *p = *p + *q;
```

```
    printf("%d %d %d", *p, a, *q);
```

```
    return 0;
```

```
}
```



$$\begin{aligned} *p &= *p + *q \\ &= 1 + 1 \\ &= 2 \end{aligned}$$

2 2 2

2. What code should be at \$\$\$ to print 9?

```
#include <stdio.h>
```

```
int sunbeam(int **q)
```

```
{
```

```
    int a = /*$$$*/;
```

```
    return a;
```

```
}
```

```
int main(void)
```

```
{
```

```
    int a = 3;
```

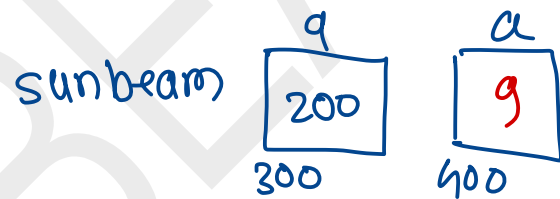
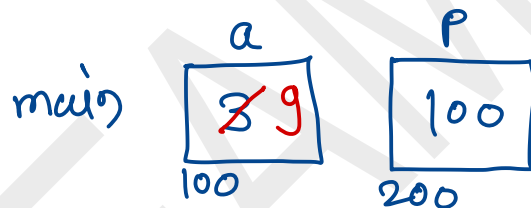
```
    int *p = &a;
```

```
    a=sunbeam(&p);
```

```
    printf("%d", a);
```

```
    return 0;
```

```
}
```



$$\begin{aligned} a &= **q * **q \\ &= 3 * 3 \\ &= 9 \end{aligned}$$

3. What will be the output of following program?

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
static int a = 1;
```

```
printf("%d", ++a);
```

```
main();
```

```
return 0;
```

```
}
```

infinite recursion

2 2 2 2

2 3 4 . . .

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4. What will be the output of following program?

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    register char reg = 20;
```

```
    char *ptr_reg = NULL;
```

```
    ptr_reg = &reg;
```

```
    *ptr_reg = 40;
```

```
    printf("%d ", reg);
```

```
    return 0;
```

```
}
```

compile time error

← register storage class variable
(can not use '&')

5. What will be the output of following program?

```
#include <stdio.h>
```

```
int var = 1;
```

```
int main(void)
```

```
{
```

```
    auto int var, var1;
```

```
    var = init();
```

```
    for(var1 = 1 ; var1 <= 3 ; var1++)
```

```
    {
```

```
        printf("%d ", first(var));
```

```
        printf("%d ", last(var));
```

```
    }
```

```
    return 0;
```

```
}
```

```
int init(void)
```

```
{
```

```
    return (var);
```

```
}
```

```
int first(int var)
```

```
{
```

```
    return var = var++;
```

```
}
```

```
int last(int var1)
```

```
{
```

```
    static int var = 5;
```

```
    return var1 = var--;
```

```
}
```

var



var



var1



main

var1 <= 3

1, 2, 3, 4

X

1 5 | 4 | 3

return (1)

var



return (1)

var1



var



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1. What will be the output of following program?

```
#include <stdio.h>
#define area(x) (x * x)
int main(void)
{
    printf("%d ", area(6 + 3));
    return 0;
}
```

$$(6 + 3 * 6 + 3) = 27$$

2. What will be the output of following program?

```
#include <stdio.h>
#define print_mul(x, y) printf("%d * %d = %d", x, y, x*y)
int main(void)
{
    int a = 3;
    print_mul(a, 3);
    return 0;
}
```

$$\text{printf}("%d * %d = %d", a, 3, a*3);$$

$$3 * 3 = 9$$

3. What will be the output of following program?

```
#include <stdio.h>
#define greater(a, b) (a > b)? a : b
int main(void)
{
    int x = 3, y = 4;
    if(greater(x, y))
        printf("sun");
    else
        printf("beam");
    return 0;
}
```

$$(x > y) ? x : y$$

Sun

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4. What will be the output of following program?

```
#include <stdio.h>
#define max(a, b) (a > b)? a : b
int main()
{
    int x = 3, y = 4, z;
    z = max(++x, y++);
    printf("%d, %d, %d", x, y, z);
    return 0;
}
```



$z = (++x > y++) ? ++x : y++;$

$z = 4 > 4 ? : 5$

4, 6, 5

5. What will be the output of following program?

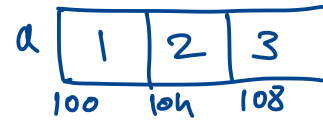
```
#include<stdio.h>
#define int char
char int main( void )
{
    char int* i = 65, j=56;
    printf("sizeof(i)=%d sizeof(j)=%d",sizeof(i),sizeof(j));
    return 0;
}
```

$\text{sizeof}(i) = 8 \quad \text{sizeof}(j) = 1$

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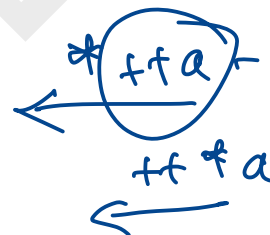
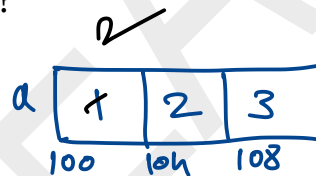
1. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    int a[] = {1, 2, 3};
    printf("%d, %d, ", sizeof(a), sizeof(a[-1]));
    print_size(a);
    return 0;
}
int print_size(int a[])  $\rightarrow$   $\text{int}^* a$ 
{
    printf("%d, %d, ", sizeof(a), sizeof(a[3]));
    return 0;
}
```



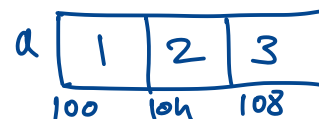
2. What will be the output of following program?

```
#include <stdio.h>
int main()
{
    int a[] = {1, 2, 3};
    ++a;  $\leftarrow$  compile time error
    printf("%d", a[2]);
    return 0;
}
```



3. What will be the output of following program?

```
#include <stdio.h>
int main()
{
    int a[] = {1, 2, 3};
    f(a);
    return 0;
}
int f(int a[])
{
    ++a;
    printf("%d", a[-1]);
    return 0;
}
```

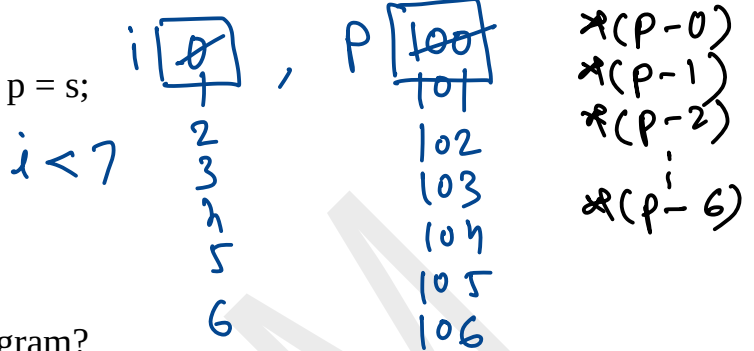
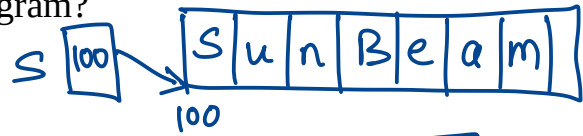


$$\begin{aligned}
 a[-1] &= *(a-1) \\
 &= *(104-1) \\
 &= *100 \\
 &= 1
 \end{aligned}$$

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4. What will be the output of following program?

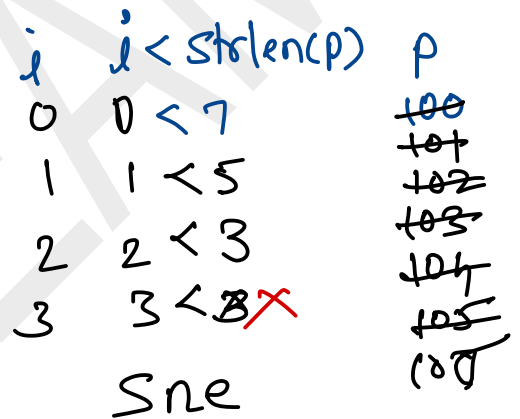
```
#include <stdio.h>
#include <string.h>
int main()
{
    char *s = "SunBeam"; int i; char *p = s;
    for(i = 0; i < strlen(s); ++i, ++p)
        printf("%c", p[-i]);
    return 0;
}
```



5. What will be the output of following program?

```
#include <stdio.h>
#include <string.h>
int main()
{
    char *s = "SunBeam"; int i; char *p = s;
    for(i = 0; i < strlen(p); ++i, ++p)
        printf("%c", *p++);
    return 0;
}
```

SSSSSSSS



6. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    char *s = "SunBeam"; char p[7] = "SunBeam";
    printf("%d, %d, %d, %d", sizeof(s), sizeof(p), sizeof(*p),
    sizeof("sunbeam"));
    return 0;
}
```

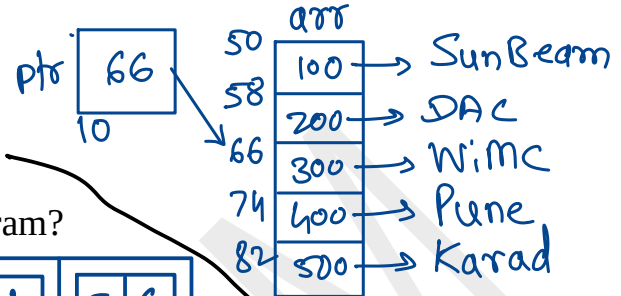
8 7 1

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1. What will be the output of following program?

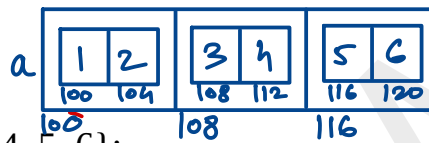
```
#include <stdio.h>
int main(void)
{
    char *arr[] = {"SunBeam", "DAC", "WiMC", "Pune", "Karad"};
    char **ptr = arr + 2;
    printf("%s", ++ptr[arr-ptr] - 1);
    return 0;
}
```

$arr[i] = *(arr+i)$



2. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    int a[][2] = { 1, 2, 3, 4, 5, 6 };
    int *p = (int *)a;
    printf("%d", p[5] - p[1]);
    return 0;
}
```

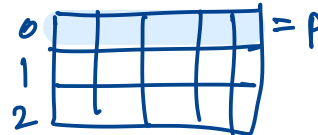


$p[5] - p[1]$
 $*(100+5) - *(100+1)$
 $*120 - *104$
 $6 - 2 = 4$

$++ptr[50-66]-1$
 $++ptr[-2]-1 = 100$
 $ptr[-2] = *(ptr-2)$
 $= *(66-2)$
 $= 100$

3. What will be the output of following program?

```
#include <stdio.h>
int main()
{
    char s[3][128] = {"SunBeam", "DAC", "WIMC"};
    char *p = s[1];
    s[0] = p; ← compile time error
    printf("%s", s[1]);
    return 0;
}
```



4. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    char s[3][128] = {"SunBeam", "DAC", "WIMC"};
    char *p = (char *)s[0];
    printf("%s", s[++p - s]);
    return 0;
}
```

← compile time error

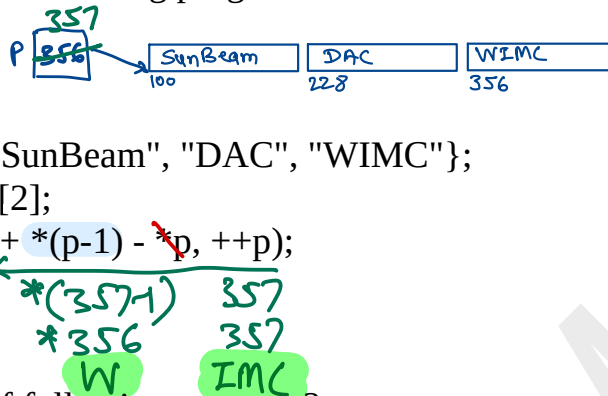


$s[++100 - 100]$
 base addr of 1st array base addr of 2D array
 Sunbeam Institute of Information Technology, Hinjawadi

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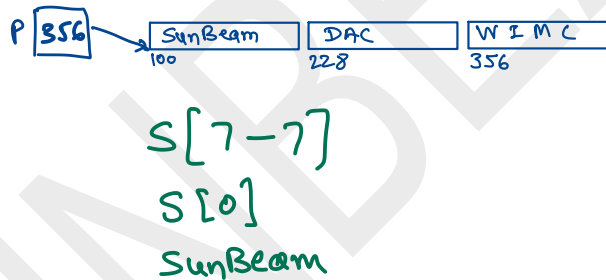
5. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    char s[3][128] = {"SunBeam", "DAC", "WIMC"};
    char *p = (char *)s[2];
    printf("%c%s", *p + *(p-1) - *p, ++p);
    return 0;
}
```



6. What will be the output of following program?

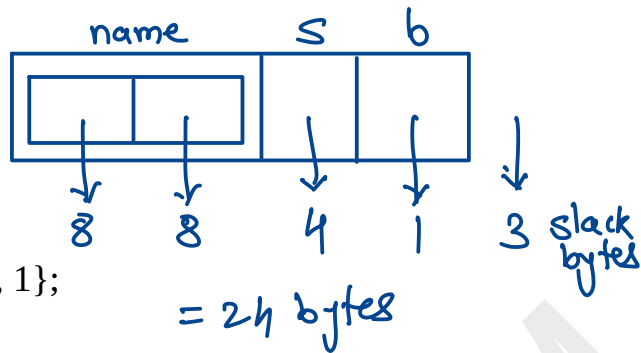
```
#include <stdio.h>
int main(void)
{
    char s[3][128] = {"SunBeam", "DAC", "W I M C"};
    char *p = (char *)s[2];
    printf("%s", s[strlen(p) - strlen(s)]);
    return 0;
}
```



PG-DESD - C Programming Twisters Activity

1. What will be the output of following program?

```
#include <stdio.h>
struct student
{
    char *name[2];
    int s;
    char b;
} s = {"Sun", "Beam", 0, 1};
int main()
{
    printf("%d", sizeof(s));
    return 0;
}
```

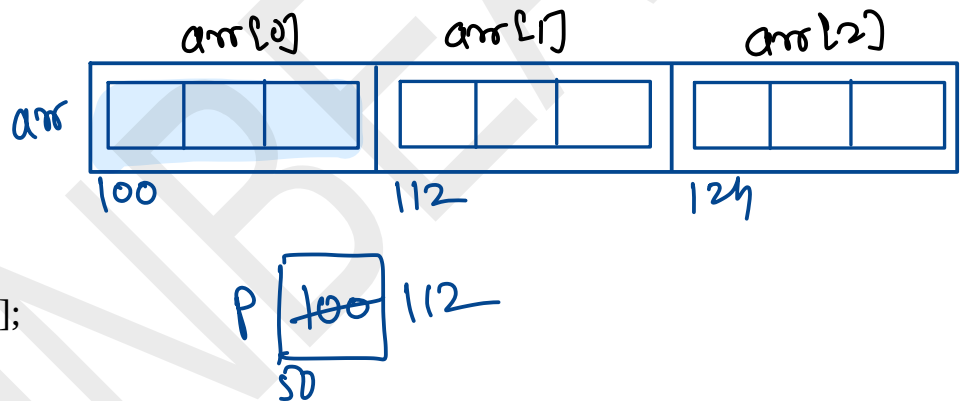


struct test {
 .
} t;
↳ variable

typedef test {
 .
} t;
↳ data type

2. What will be the output of following program?

```
#include <stdio.h>
struct x
{
    int i; int j; int k;
};
int main(void)
{
    struct x *p, arr[3];
    p = &arr[0];
    ++p;
    printf("%d", &(arr[0].j) - p);
    return 0;
}
```



int * ← ↑ → struct x *

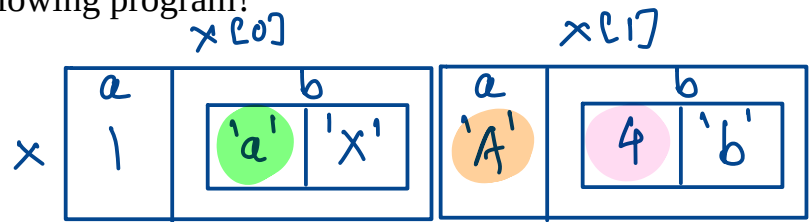
compile time error

PG-DESD - C Programming Twisters Activity

3. What will be the output of following program?

```
#include <stdio.h>
int main(void)
{
    struct a
    {
        int a;
        struct b
        {
            int a; char b;
        } b;
    } x[] = {1, 'a', 'x', 'A', 4, 'b'};

    printf("%d", -x[1].a + x[0].b.a + x[1].b.a);
    return 0;
}
```

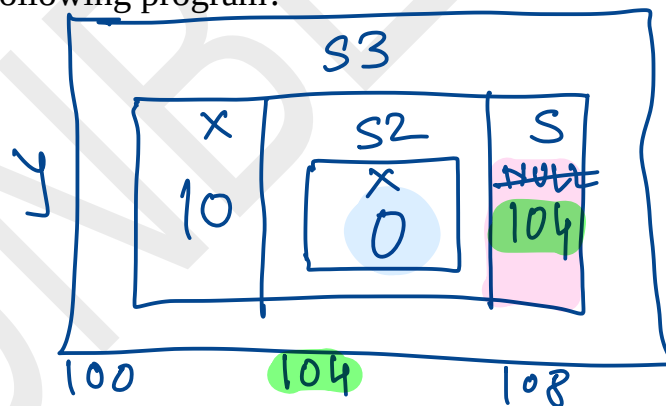


$$-65 + 97 + 4 = 36$$

4. What will be the output of following program?

```
#include <stdio.h>
struct s1
{
    struct {
        int x;
        struct s { int x;}s2;
        struct s *s;}s3;
}y={ 10 };

int main(void)
{
    y.s3.s = &y.s3.s2;
    printf("%d", y.s3.s->x);
    return 0;
}
```



PG-DESD - C Programming Twisters Activity

1. What will be the output of following program?

```
#include<stdio.h>
enum color {red,green,yellow=1,dark};
int main(void)
{
    enum color x = green % dark;
    printf("%d", yellow + dark/x - green * red);
    return 0;
}
```

Handwritten annotations:
 yellow = 1, dark = 2
 $1 + 2/1 - 1 * 0 = 3$

2. What will be the output of following program?

```
#include<stdio.h>
int main(void)
{
    enum choice {CH1, CH2, CH3};
    enum choice ch1, ch2, ch3;
    ch1 = CH1;
    ch2 = CH3;
    ch3 = CH2;
    printf("%d, %d, %d", ch1, ch2, ch3);
    return 0;
}
```

Handwritten annotations:
 CH1 = 0, CH2 = 1, CH3 = 2
 ch1 = 0, ch2 = 2, ch3 = 1

3. What will be the output of following program?

```
#include<stdio.h>
int main(void)
{
    enum choice {CH1, CH2, CH3};
    enum choice ch1, ch2, ch3;
    ch1 = CH1;
    ch2 = CH3;
    ch3 = ch2 - ch1;
    printf("%d %d", sizeof(enum choice), ch3);
    return 0;
}
```

Handwritten annotations:
 CH1 = 0, CH2 = 1, CH3 = 2
 ch1 = 0, ch2 = 2, ch3 = 2 - 0 = 2
 sizeof(enum choice) = 4

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4. What type of data u can store in this block of memory?

```
#include<stdio.h>
#include<stdlib.h>
int main(void)
{
    void *ptr=NULL;
    ptr = malloc(10);
    return 0;
}
```

→ Any

5. What will be the output of following program?

```
#include<stdio.h>
#include<stdlib.h>
int main(void)
{
    char *ptr=NULL;
    ptr = (char *)calloc(20,sizeof(char));
    printf("%d bytes\n", sizeof(ptr));
    free(ptr); ptr=NULL;
    return 0;
}
```

↳ 8 bytes

6. What will be the output of following program?

```
#include<stdio.h>
#include<stdlib.h>
int main(void)
{
    char *ptr=NULL;
    ptr = (char *)calloc(1,10);
    strcpy(ptr, "Sunbeam");
    ptr = (char *)realloc(ptr,20);
    strcat(ptr," IT PARK");
    printf("%c", (ptr[0]>=65 && ptr[0]<=90)? ptr[14]+32 : ptr[0]-32);
    free(ptr);
    ptr=NULL;
    return 0;
}
```

Sunbeam IT PARK
 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

S >= 65 && S <= 90
 ✓ ✓

↳ 'K' + 32

= 'k'

PG-DESD - C Programming Twisters Activity

5. What will be the output of following program?

```
#include<stdio.h>
union u
{
    int x[3];
    char y[5];
}u[3];

int main()
{
    printf("%d", sizeof(u));
    return 0;
}
```

6. What will be the output of following program?

```
#include<stdio.h>
struct s
{
    int x[10];
};
union u
{
    int *x;
    struct s *y[2];
}u;
int main(void)
{
    printf("%d", sizeof(u));
    return 0;
}
```

PG-DESD - C Programming Twisters Activity

1. What will be output on 32-bit compiler?

```
#include <stdio.h>
int main()
{
    int a = 10;
    unsigned b;
    b = sizeof(++a + a++ + ++a) + a;
    printf("%d %d\n", a, b);
    return 0;
}
```

- A. 13 14
- B. 10 14
- C. 13 17
- D. None of these

2. If input user inputs 10 & 100, then what will be output?

```
#include <stdio.h>
int main()
{
    int a, b;
    printf("%d %d\n", printf("%d %d ", a, b), scanf("%d%d", &a, &b));
    return 0;
}
```

- A. 10 100 7 2
- B. Garbage Garbage Garbage 2
- C. 10 100 2 2
- D. None of these

3. What will be the output of following program?

```
#include <stdio.h>
int main()
{
    int a = 0, b = 3, c = 6, d;
    d = a++ && ++b || c++;
    printf("%d %d %d %d\n", a, b, c, d);
    return 0;
}
```

- A. 1 4 7 1
- B. 1 4 7 0
- C. 1 3 7 1
- D. None of these

PG-DESD - C Programming Twisters Activity

4. What is the value printed by the following C program?

```
int main()
{
    int a=3,b=4,c;
    c=a+b/a*a*a;
    printf("%d\n",c);
    return 0;
}
```

- A. 3
- B. 12
- C. 11
- D. 8

5. What will be the output of following program?

```
#include<stdio.h>
int main(void)
{
    int x = 10;
    (x < 0) ? int x = 100 : {int x = 1000};
    printf("%d", x);
    return 0;
}
```

- A. 1000
- B. 100
- C. 10
- D. Compile time error

6. What will be the output of following program?

```
#include<stdio.h>
int main(void)
{
    int a=3, b=5, c=50;
    a=b==c;
    printf("%d",a);
}
```

- A. 3
- B. 5
- C. 50
- D. 0

PG-DESD - C Programming Twisters Activity

7. What will be the output of following program?

```
#include<stdio.h>
int main(void)
{
    int a=14;
    a += 7;
    a -= 5;
    a *= 7;
    a ^= a;
    printf("%d", a);
    return 0;
}
```

- A. 112
- B. 98
- C. 89
- D. None of these